

CS & IT ENGINEERING



Theory of Computation

DFA

Lecture No.- 07



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Recap of Previous Lecture



Topic

DFA

Topic



(Q) Construct min state DFA over $\Sigma = \{a, b\}$ where third input symbol is a from R.H.S.

8th {256} ✓
R.H.S

3rd → 8

4 → 2⁴

5 → 2⁵

nth → (2ⁿ) states

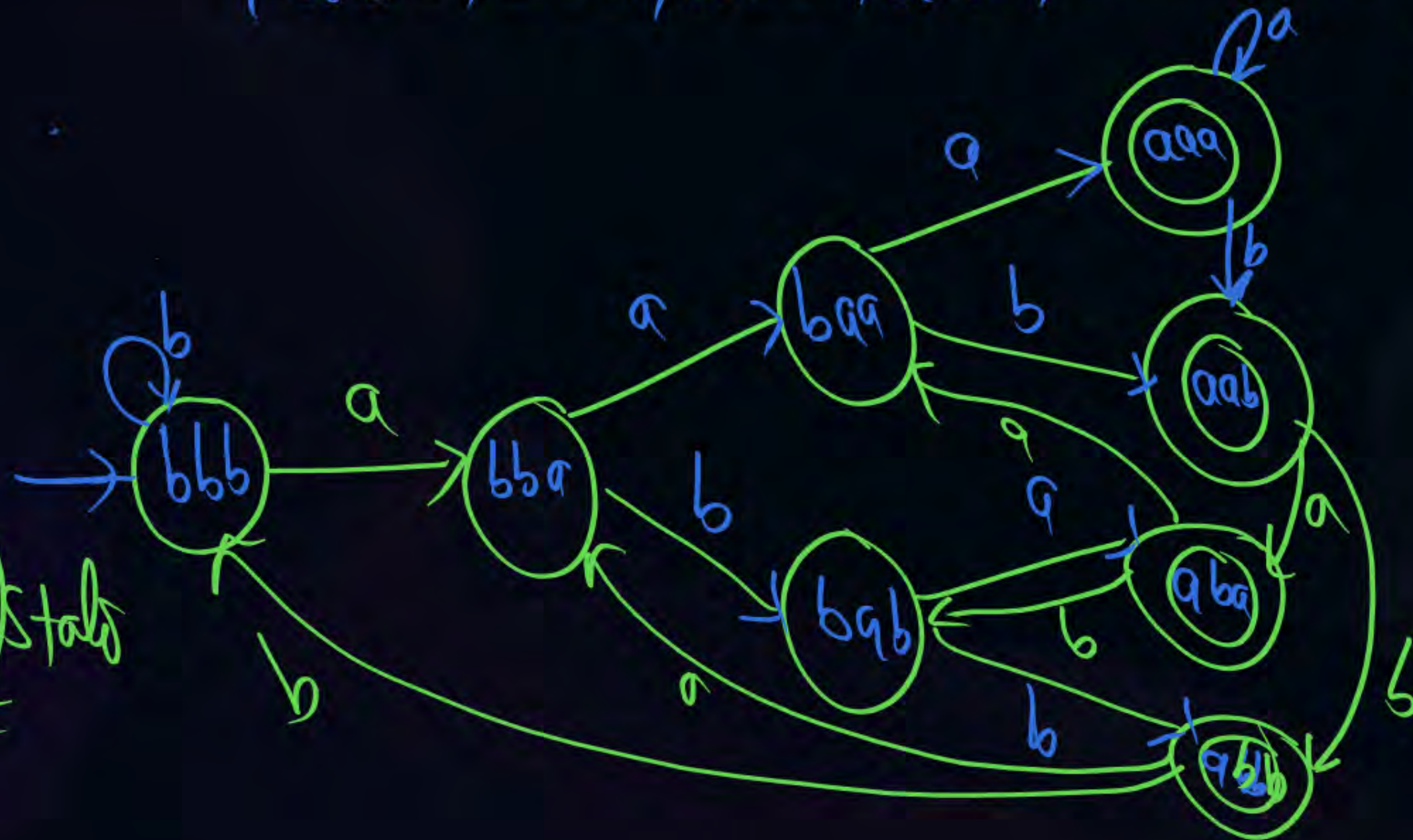
(a/b) a a/b a/b

{abb, aba, aaa, aab, aaba, ...}

Possible daab
aaba

(NFA)
1 min

aabb
abaa
abab
abba
abbb



Q Construct NFA over $\Sigma = \{a, b\}$ where 3rd input is a from R.H.S.



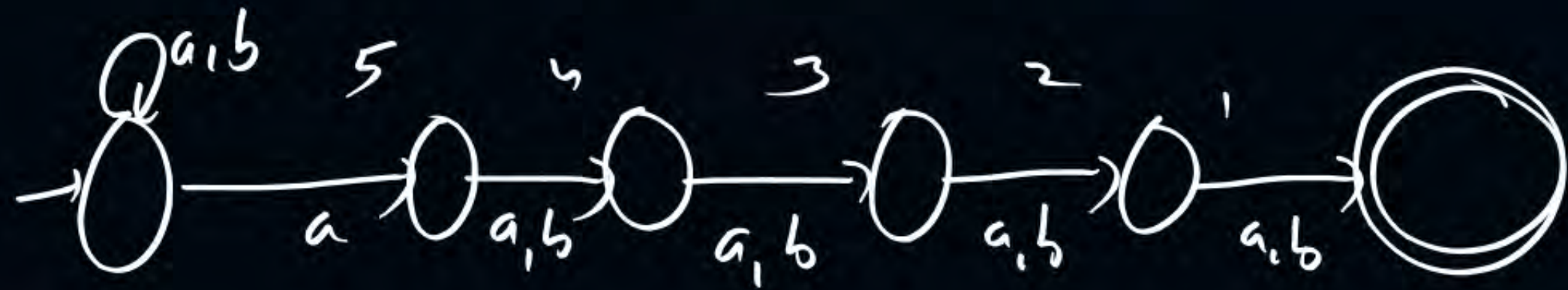
$\begin{pmatrix} a \\ b \end{pmatrix}$ $\xrightarrow[3]{a}$ $\xrightarrow[2]{a/b}$ $\xrightarrow[1]{a/b}$



Q Construct NFA over $\Sigma = \{a, b\}$ where 5th input is a from R.H.S.

DFA = 32

(a)
b (a) a/b a/b a/b a/b



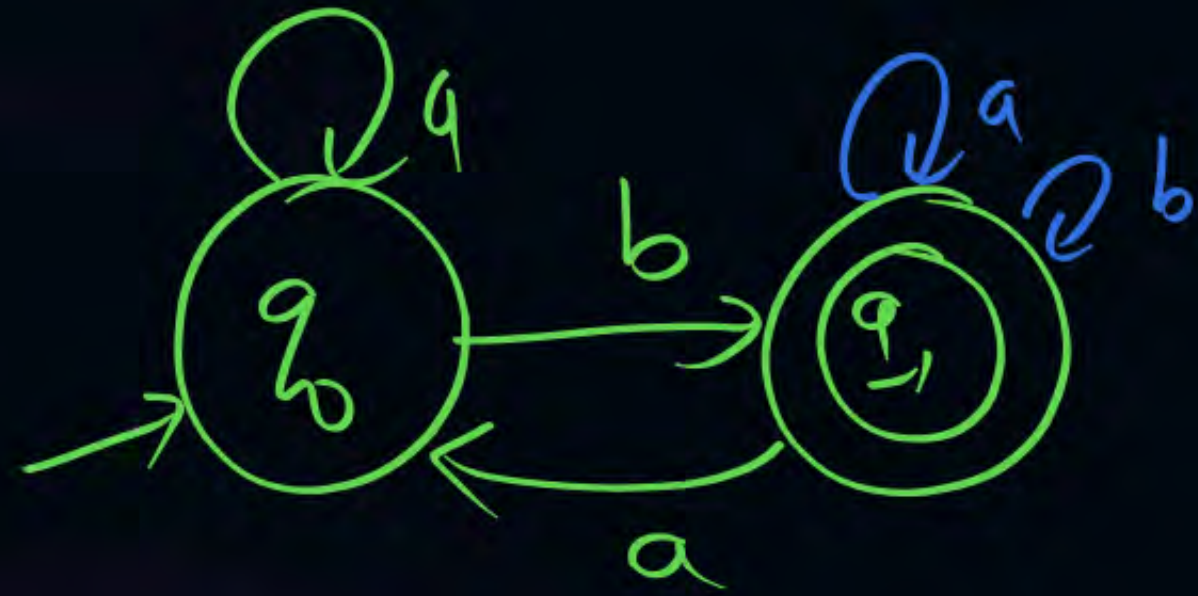
27

(n+1) states

NFA = 6

Q Construct NFA over $\Sigma = \{a, b\}$ where 3rd input is a from R.H.S.





NFA

$$\delta(q_1, b) = \emptyset$$

$$\delta(q_1, a) = \{ \underline{q_0}, \underline{q_1} \}$$



Topic : Non-Deterministic Finite Automaton

In NFA from the given state on the given input symbol there may be 0 number of transition or 1 transition or more than one transition exist.

NFA is formally define as:-

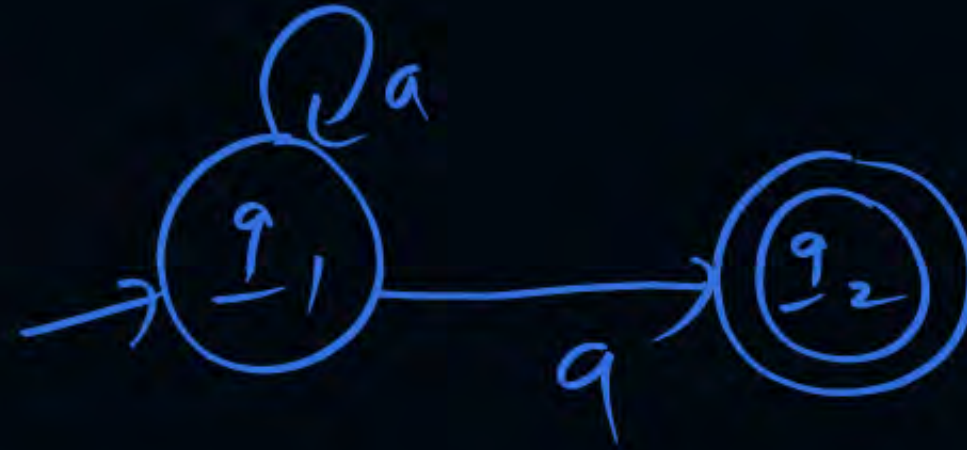
$$\text{NFA} = \{Q, \Sigma, q_0, F, \delta\}$$

- ✓ Q - Finite number of states (set of state)
- ✓ Σ - Input alphabet
- ✓ q_0 - initial state
- F - Set of final states
- ✓ δ - transition function

$$\text{DFA} = \boxed{Q \times \Sigma \rightarrow Q}$$

$$Q \times \Sigma \rightarrow 2^Q \text{ or power set}$$

NFA



$$S = \{q_1, q_2\}$$

$$P(S) = \underline{4}$$

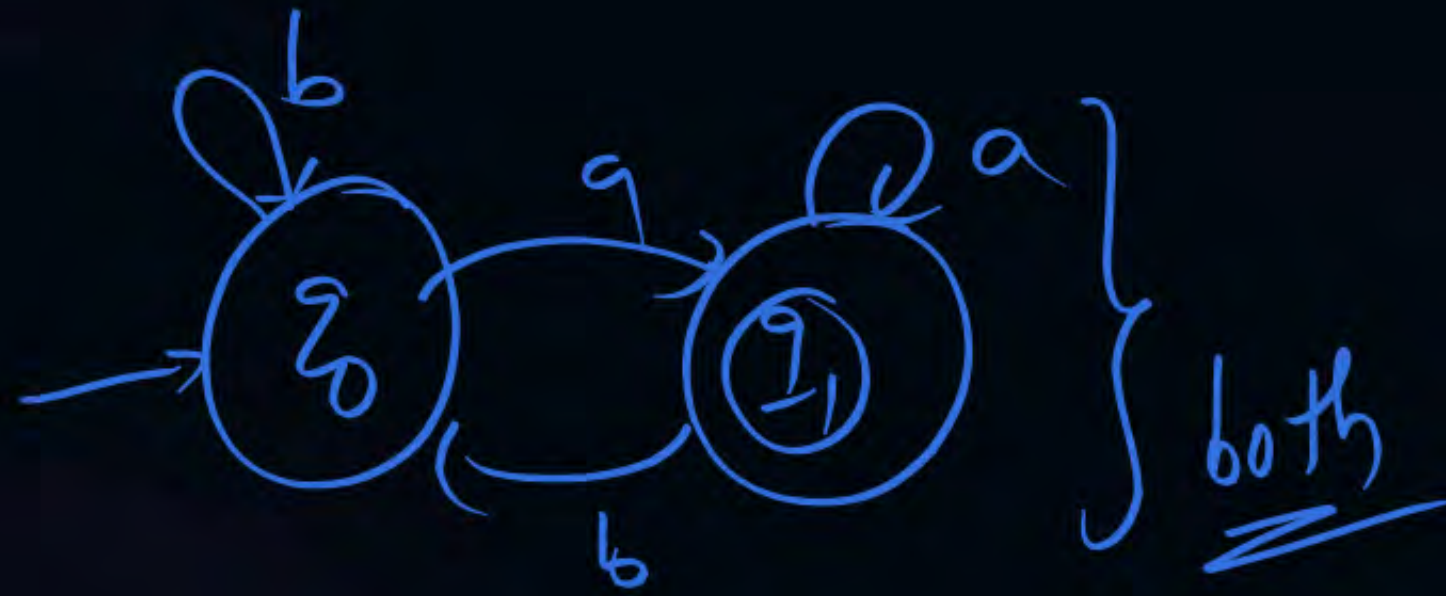
$$\delta(q_1, a) = \{ \} \quad \{q_1\} \quad \{q_2\} \quad \{q_1, q_2\}$$

$$\text{powerset}(\underline{\text{states}}) = 2^Q$$



Topic : Non-Deterministic Finite Automaton

- ① Construction of NFA is easy than DFA.
- ② Minimization not possible for NFA
- ③ Complementation not possible for NFA
- ④ NFA from the given state on the given input string multiple state possibility may be exist.
- ⑤ Language recognition is easy in DFA compare to NFA.
- ⑥ In NFA, for valid string also automata may halt in non-final state.
- ⑦ In NFA for the valid ^{string} even though multiple non-final transition exist for one final state transition should exit.
- ⑧ All DFA are NFA but all NFA need not be DFA.



DFA $\rightarrow$ NFA



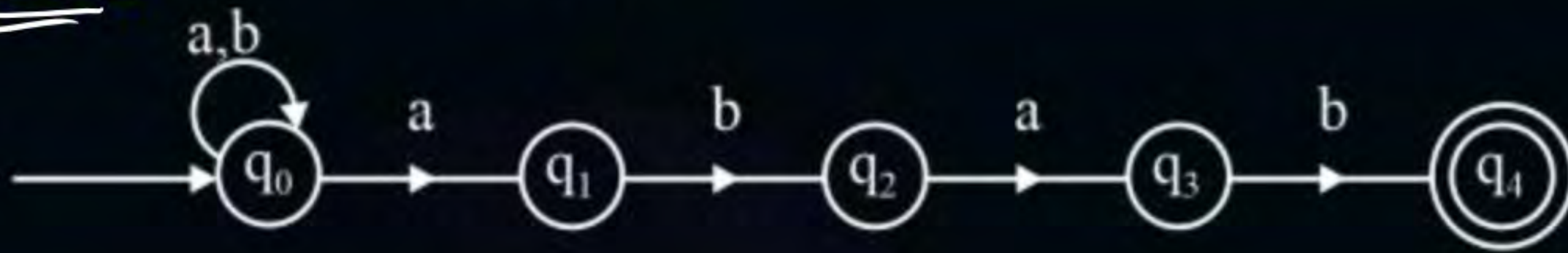
Topic : NFA

Construct the NFA that accept all strings of a's & b where every string ending with

abab.

NFA:

$\left(\begin{smallmatrix} a \\ b \end{smallmatrix} \right) \underline{abab}$

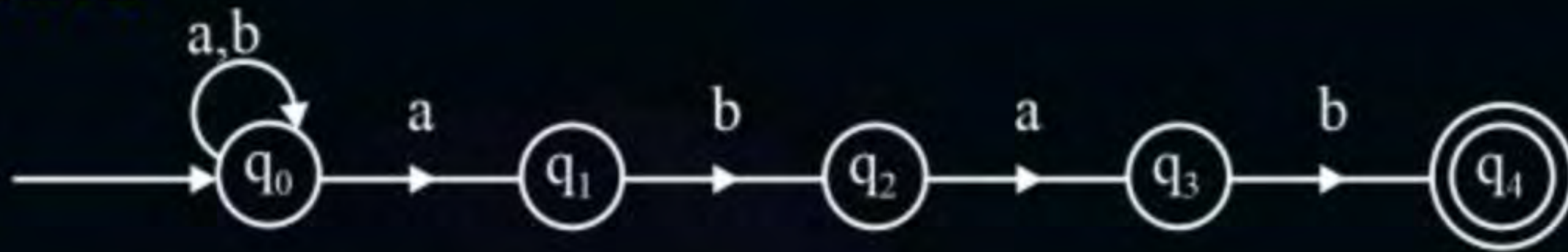




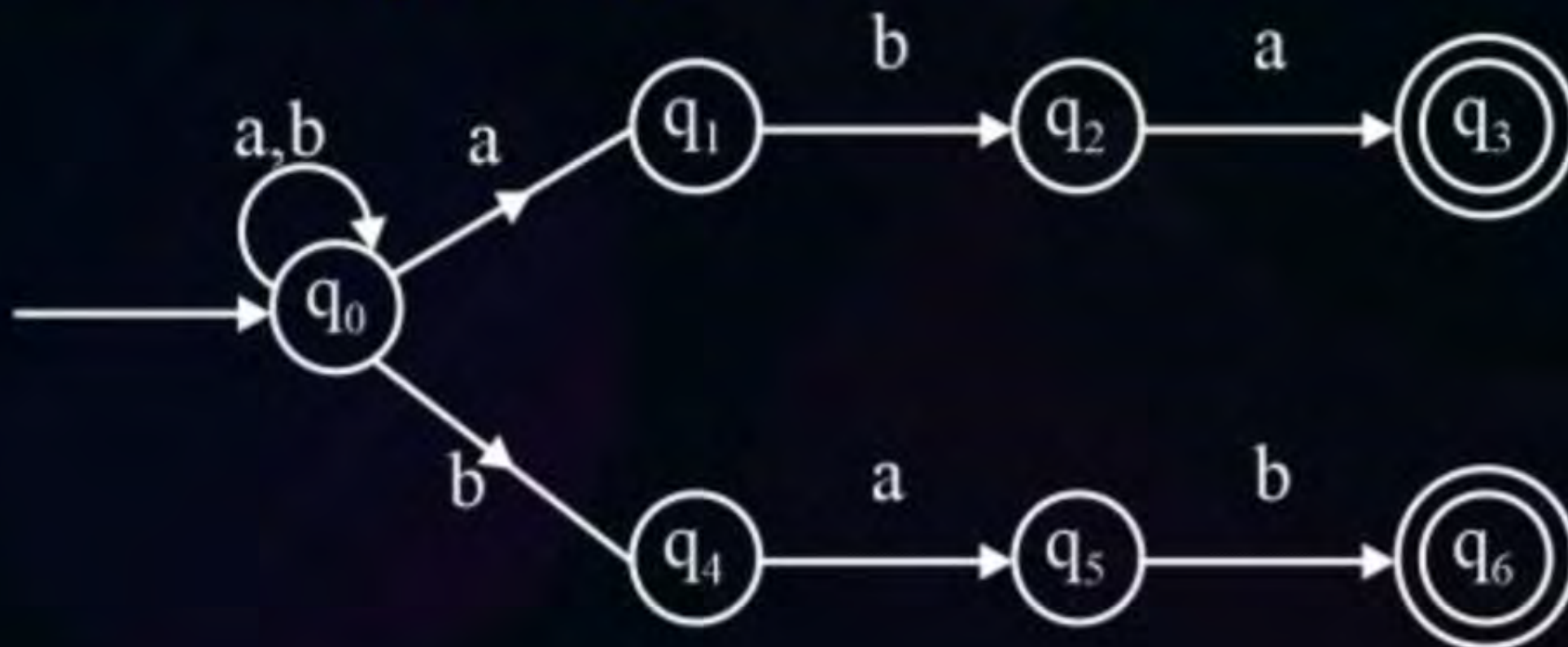
Topic : NFA

Construct the NFA that accept all strings of a's & b where every string ending with abab.

NFA:

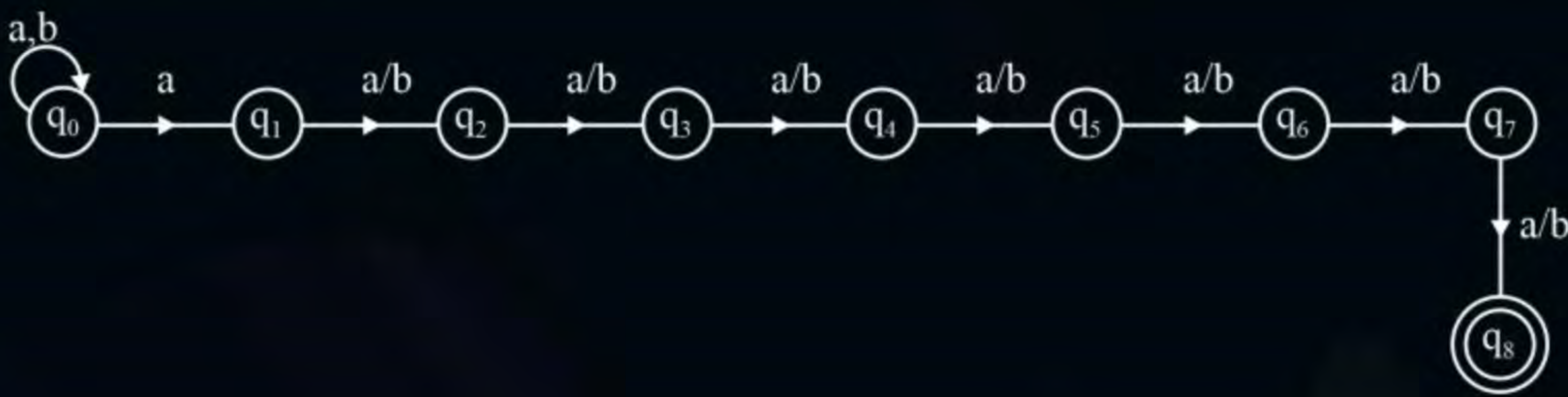


Each string ending with aba or bab





Topic : NFA



8th input symbol is a while reading from right side



Topic : NFA

Note:- The NFA that accept strings of a's & b's where n^{th} input symbol is a while reading a strings from w right hand side requires $(n+1)$ states.

Expressive Power of Automata

Number of languages accepted by particular automata is known as expressive power.

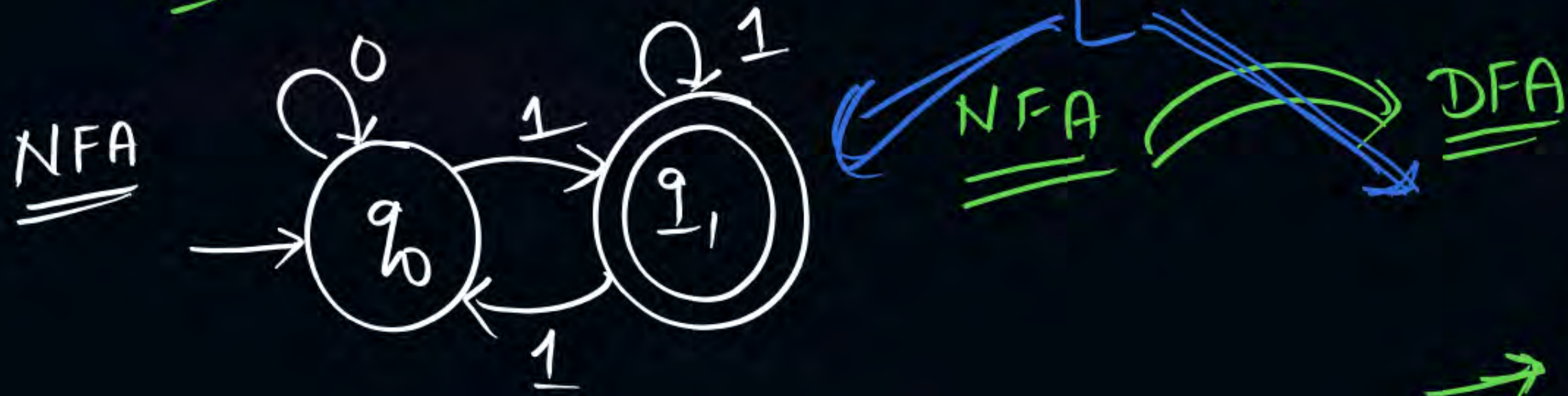


Sub set Construction Algorithm

NFA $\Rightarrow$ DFA



Q) Construct DFA for given NFA



NFA

	0	1
$\rightarrow q_0$	q_0	q_1 ✓
q_1	$\{q_0, q_1\}$ ✓	$\{q_0, q_1\}$ ✓

	0	1
$\rightarrow q_0$	q_0	q_1 ✓
q_1	$q_{D.S}$ ✓	$[q_0, q_1]$ ✓
$q_{D.S}$	$q_{D.S}$ ✓	$q_{D.S}$ ✓
$[q_0, q_1]$	q_0 ✓	$[q_0, q_1]$ ✓

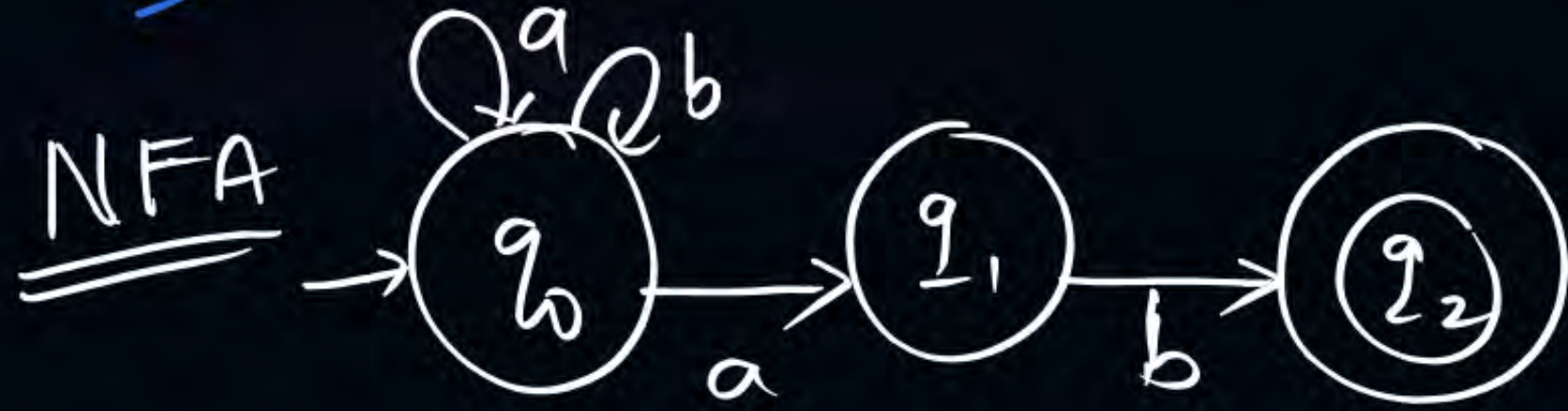
NFA $\rightarrow$ DFA

① $\emptyset \rightarrow D.S$

② $\{q_0, q_1\} \rightarrow \{\text{new state}\}$

2 final state

(1) Construct DFA for given NFA



NFA

	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	q_0
q_1	—	q_2
q_2	—	—

DFA

$\rightarrow q_0$

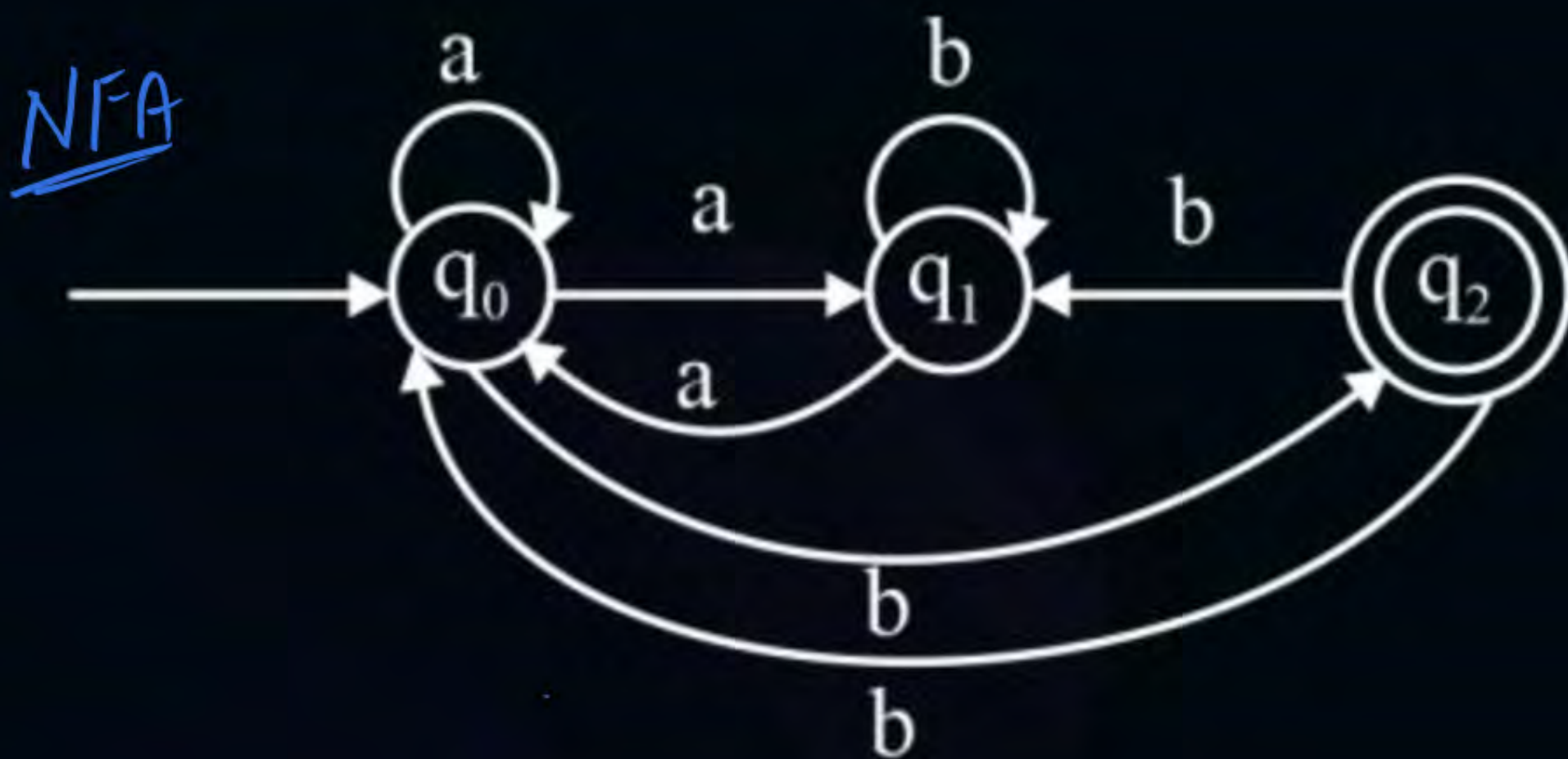
	a	b
q_0	$[q_0, q_1]$	q_0
$[q_0, q_1]$	$[q_0, q_1]$	$[q_0, q_2]$
$[q_0, q_2]$	$[q_0, q_1]$	q_0

3 states



Topic : NFA to DFA Conversion

Consider the following NFA what is the equivalent DFA for given NFA?



DFA ??

Home Work



Topic : NFA to DFA Conversion

Note:- NFA to DFA conversion Algorithm does not affect language of automata i.e. if language accept by a NFA is L then while converting that NFA into DFA the resultant DFA also accepts. *same language*

NFA:

Home work



DFA



THANK - YOU